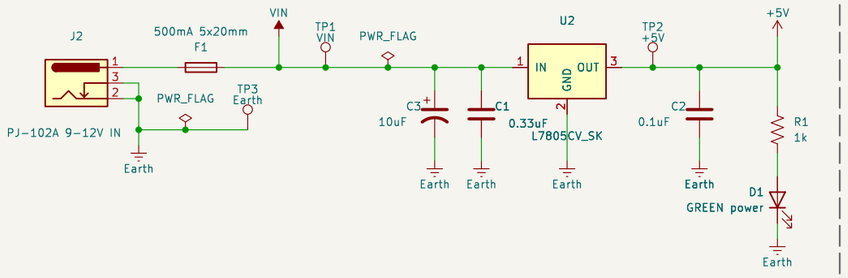
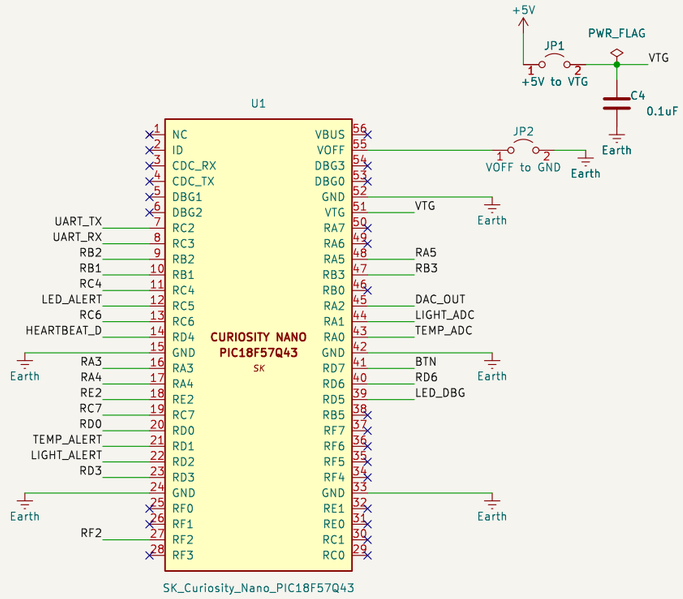
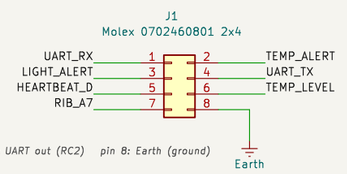
1. POWER INPUT AND 5 V LINEAR REGULATOR



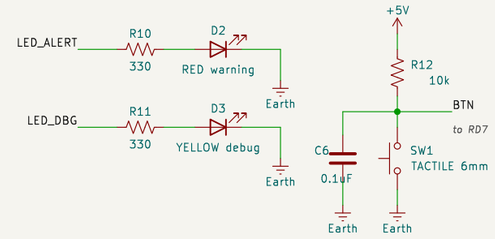
2. MICROCONTROLLER: PIC18F57Q43 CURIOSITY NANO (custom symbol)



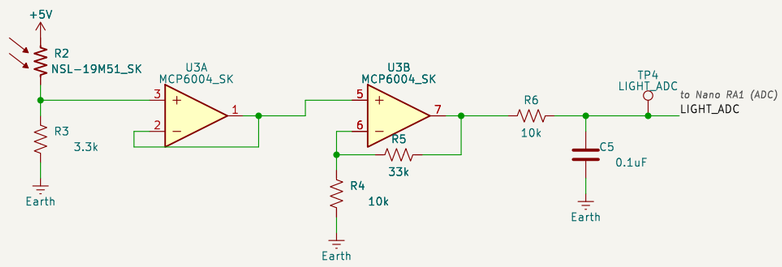
3. TEAM RIBBON CABLE J1 (2x4 IDC, course standard)



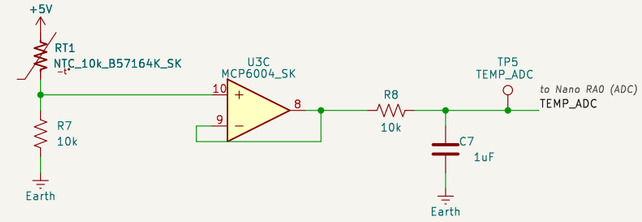
4. DEBUG LEDs AND PUSHBUTTON



5. LIGHT SENSOR (NSL-19M51 photocell, custom symbol) WITH BUFFER, GAIN STAGE AND LOW-PASS FILTER



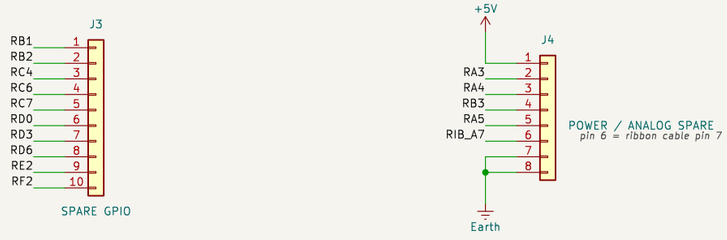
6. TEMPERATURE SENSOR (10k NTC thermistor, custom symbol) WITH BUFFER AND LOW-PASS FILTER



7. DAC OUTPUT BUFFER (TEMP_LEVEL TO TEAMMATES) AND OP-AMP POWER



8. SPARE GPIO / EXPANSION HEADER AND POWER HEADER



NOTE: all resistors 1/4 W (R7 1 % tolerance). All ceramic capacitors 50 V; C3 electrolytic 25 V.

Ribbon cable J1: pin 1 UART in (RC3 RX), pin 4 UART out (RC2 TX), pin 8 Earth (ground)
PIC18F57Q43 Curiosity Nano, 5 V rail from L7805CV, MCP6004 signal conditioning
Sam Kholmuminov (SK)
Team 102 — Project Aurora, EGR 304

Sheet: /
File: Storage_Environment_Board_SK.kicad_sch

Title: Storage Environment Board (Board D) — Schematic

Size: A3	Date: 2026-10-20	**Rev: 3**
KiCad E.D.A. kicad-cli 7.0.11+dfsg-1build4		Id: 1/1